

Interference of Light

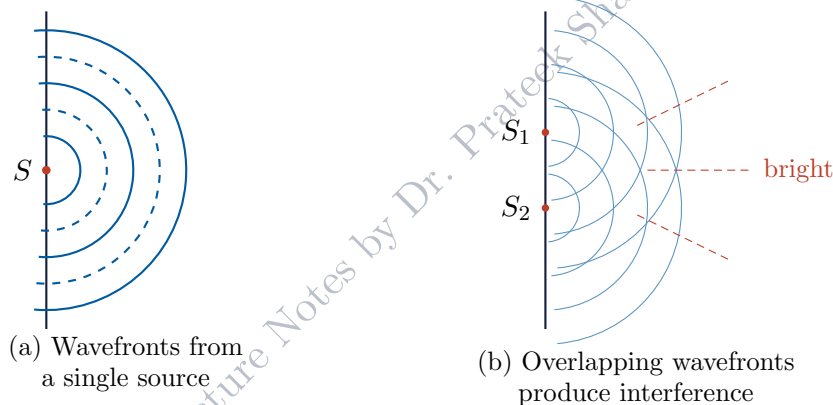
B.Sc. (V Semester) — Physical Optics · Kumaon University
Unit II: Interference · Lecture 1 · Instructor **Dr. Prateek Sharma**

Interference · Principle of superposition · Intensity · Visibility of fringes

1 Interference

The modification in the distribution of light energy due to the superposition of two or more waves is called **interference**.

When two coherent light waves superpose, the resultant intensity is redistributed: it is maximum where the waves arrive in phase (*constructive interference*) and minimum where they arrive out of phase (*destructive interference*).



Coherent sources

Two sources that vibrate with a **constant (fixed) phase difference** between them are said to be *coherent*. Only coherent sources produce a steady, observable interference pattern.

1.1 Classification by mode of superposition

Division of wavefront.

A single wavefront is divided into two parts using reflection, refraction or diffraction, so that after traversing slightly different optical paths the light from the two coherent portions produces interference bands. (e.g. *Young's slits, Fresnel's biprism*)

Division of amplitude.

The amplitude of an incoming wave is divided into two (or more) parts by partial reflection and refraction, and these beams subsequently reunite to produce interference. (e.g. *thin films, Newton's rings*)

2 Principle of Superposition

A plane wave travelling along the x -direction may be written

$$y = a \sin(\omega t - \phi),$$

where y is the instantaneous displacement, a the amplitude of the electric field vector, $\omega = 2\pi n$ the angular frequency (n = frequency), λ the wavelength and ϕ the initial phase.

Consider two waves of the same frequency arriving at a point:

$$y_1 = a_1 \sin(\omega t - \phi_1), \quad y_2 = a_2 \sin(\omega t - \phi_2).$$

By the principle of superposition the resultant displacement is $y = y_1 + y_2$. Expanding,

$$y = a_1 (\sin \omega t \cos \phi_1 - \cos \omega t \sin \phi_1) + a_2 (\sin \omega t \cos \phi_2 - \cos \omega t \sin \phi_2),$$

$$y = \sin \omega t (a_1 \cos \phi_1 + a_2 \cos \phi_2) - \cos \omega t (a_1 \sin \phi_1 + a_2 \sin \phi_2).$$

Introduce constants A and θ through

$$A \cos \theta = a_1 \cos \phi_1 + a_2 \cos \phi_2, \quad A \sin \theta = a_1 \sin \phi_1 + a_2 \sin \phi_2. \quad (1)$$

Then

$$y = A \sin \omega t \cos \theta - A \cos \omega t \sin \theta = A \sin(\omega t - \theta).$$

The resultant is a wave of the **same frequency**, with amplitude A and phase θ .

Squaring and adding the two relations in (1),

$$\begin{aligned} A^2 &= A^2 \cos^2 \theta + A^2 \sin^2 \theta \\ &= (a_1 \cos \phi_1 + a_2 \cos \phi_2)^2 + (a_1 \sin \phi_1 + a_2 \sin \phi_2)^2 \\ &= a_1^2 + a_2^2 + 2a_1 a_2 (\cos \phi_1 \cos \phi_2 + \sin \phi_1 \sin \phi_2) \\ &= a_1^2 + a_2^2 + 2a_1 a_2 \cos(\phi_2 - \phi_1). \end{aligned}$$

Writing the phase difference $\phi = \phi_2 - \phi_1$,

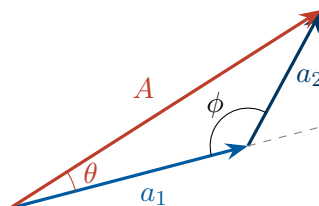
$$A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi$$

Dividing the two relations in (1) gives the phase of the resultant:

$$\tan \theta = \frac{a_1 \sin \phi_1 + a_2 \sin \phi_2}{a_1 \cos \phi_1 + a_2 \cos \phi_2}.$$

Phasor picture

The same result follows geometrically by adding the two amplitudes as vectors (phasors); the resultant A closes the triangle and its magnitude obeys the cosine rule.



3 Intensity

Since intensity is proportional to the square of the amplitude, $I = A^2$:

$$I = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi$$

The interference term $2a_1a_2 \cos \phi$ redistributes the energy.

Condition for maximum intensity

$$\cos \phi = +1 \Rightarrow \phi = 2n\pi, \quad n = 0, 1, 2, \dots$$

$$I_{\max} = a_1^2 + a_2^2 + 2a_1a_2 = (a_1 + a_2)^2$$

Condition for minimum intensity

$$\cos \phi = -1 \Rightarrow \phi = (2n + 1)\pi, \quad n = 0, 1, 2, \dots$$

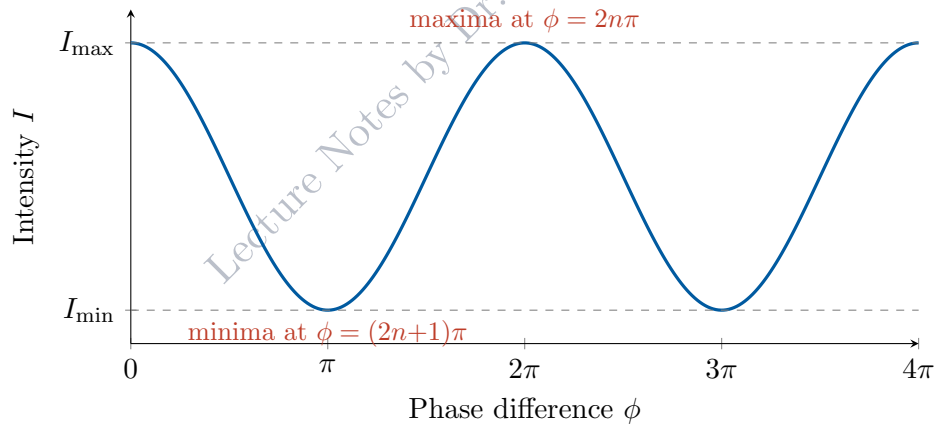
$$I_{\min} = a_1^2 + a_2^2 - 2a_1a_2 = (a_1 - a_2)^2$$

Equal amplitudes

When $a_1 = a_2 = a$,

$$I_{\max} = 4a^2, \quad I_{\min} = 0.$$

The minima are perfectly dark only when the two amplitudes are equal.



(Intensity vs. phase difference, drawn for $a_1 = 2a_2$; here $I_{\min} \neq 0$ because the amplitudes are unequal.)

4 Visibility of Fringes (Fringe Contrast)

The sharpness of the pattern is measured by the visibility, or fringe contrast:

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

Using $I_{\max} = (a_1 + a_2)^2$ and $I_{\min} = (a_1 - a_2)^2$:

$$I_{\max} - I_{\min} = (a_1 + a_2)^2 - (a_1 - a_2)^2 = 4a_1a_2,$$

$$I_{\max} + I_{\min} = (a_1 + a_2)^2 + (a_1 - a_2)^2 = 2(a_1^2 + a_2^2).$$

Therefore

$$V = \frac{4a_1a_2}{2(a_1^2 + a_2^2)} = \frac{2a_1a_2}{a_1^2 + a_2^2}.$$

Equal amplitudes ($a_1 = a_2$):

$$V = \frac{2a^2}{2a^2} = 1 \quad (\text{maximum possible contrast}).$$

Unequal amplitudes. Let $I_1 \propto a_1^2$ and $I_2 \propto a_2^2$ with $\frac{I_2}{I_1} = \frac{a_2^2}{a_1^2} = \alpha$, so that $a_2 = a_1\sqrt{\alpha}$. Then

$$V = \frac{2a_1a_2}{a_1^2 + a_2^2} = \frac{2a_1(a_1\sqrt{\alpha})}{a_1^2 + \alpha a_1^2} = \frac{2\sqrt{\alpha}}{1 + \alpha}.$$

Contrast is best when the intensities (and hence amplitudes) of the two beams are nearly equal.

5 Worked Problem

Problem. The amplitudes of the light waves emerging from the two slits in Young's experiment are in the ratio 1:2. Find the intensity ratio of the interference pattern.

Solution. Let $a_2 = 2a_1$. Then

$$\begin{aligned} I_{\max} &= a_1^2 + a_2^2 + 2a_1a_2 = a_1^2 + 4a_1^2 + 4a_1^2 = 9a_1^2, \\ I_{\min} &= a_1^2 + a_2^2 - 2a_1a_2 = a_1^2 + 4a_1^2 - 4a_1^2 = a_1^2. \end{aligned}$$

$$\frac{I_{\text{bright}}}{I_{\text{dark}}} = \frac{9a_1^2}{a_1^2} = 9.$$

Intensity ratio = 9:1